Remainders modulo 16 for the series $A(xA - 4xA^2) = x^2$

The correct form of a conjecture in OEIS A396844

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Abstract

Entry A396844 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna, lists the coefficients $a(n)$ of the power series $A(x) = x + 4x^2 + 28x^3 + 288x^4 + \cdots$ determined by the equation $A\big(xA(x) - 4xA(x)^2\big) = x^2$. The entry conjectures that, for $n > 8$, the remainder of $a(n)$ on division by 16 is 12 when $n = 2 \cdot 4^k + 1$, and 0 otherwise. We show that the first half of this statement is true and the second half is not: for every $n = 4^k + 1$ the remainder is 4, the first exception after $n = 8$ being $n = 17$. The complete rule, which we prove, is: for $n \geq 2$, the remainder is 4 if $n = 4^k + 1$, it is 12 if $n = 2 \cdot 4^k + 1$, and it is 0 otherwise. We also show that $a(n)$ is divisible by 4^s, where s is the number of ones in the binary expansion of $n - 1$. Both results hold, in a form that depends on m, for the whole family of equations $A\big(xA - mxA^2\big) = x^2$. The results were checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [7] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A396844 [4], contributed by Paul D. Hanna in July 2026. It is a companion of entry A396843 [3]. The entry defines a power series

$$A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots = x + 4x^2 + 28x^3 + 288x^4 + 3204x^5 + \cdots$$

by a single equation,

$$A(x\,A(x) - 4\,x\,A(x)^2) = x^2. \tag{1}$$

To read the left side, first form the series $T(x) = x\,A(x) - 4\,x\,A(x)^2$, which begins with x^2, and then substitute $T(x)$ for x in A. (Section 2 explains what substituting one series into another means.) The equation says that the result is simply x^2. The entry reads:

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G.f.  A(x) satisfies A( x*A(x) - 4*x*A(x)^2 ) = x^2.

Conjecture:  for n > 8, a(n) == 12 (mod 16) iff n is of the form 2*4^k+1
otherwise a(n) is divisible by 16.
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— OEIS A396844, Paul D. Hanna, July 2026

Here "G.f." stands for *generating function*, the series whose coefficients are the terms of the sequence.

It is not obvious that equation (1) pins down a series at all, so let us see how it determines the coefficients. Write $A = x + a(2)x^2 + \cdots$, with first coefficient $a(1) = 1$ as in the entry. Then

$$T = x\,A - 4\,x\,A^2 = x^2 + (a(2) - 4)x^3 + \cdots, \qquad A(T) = T + a(2)\,T^2 + \cdots = x^2 + (a(2) - 4)x^3 + \cdots,$$

because T^2 begins with x^4. For this to equal x^2, the coefficient of x^3 must vanish, so $a(2) = 4$. In the same way, the coefficient of x^{n+1} in (1) determines $a(n)$ once $a(1), \ldots, a(n-1)$ are known (Lemma 4.1). So there is exactly one solution, and all its coefficients are integers. Its first terms are 1, 4, 28, 288, 3204, 37984, 470864, 6031872, 79213292, ... (starting from $n = 1$).

1.2 The question

The conjecture is about the remainders of $a(n)$ on division by 16. (Section 2 recalls the notation $a \equiv b \pmod{m}$ for "a and b leave the same remainder on division by m".) It says that from $n = 9$ on, the remainder is 12 when $n = 2 \cdot 4^k + 1$ (that is, $n = 9, 33, 129, \ldots$), and 0 for every other n.

The table shows the remainders of the first terms on division by 16. The indices in bold are those of the form $n = 2^j + 1$, namely $2, 3, 5, 9, 17, 33$.

n	1	**2**	**3**	4	**5**	6	7	8	**9**	10	11	12	13	14	15	16	**17**
$a(n) \bmod 16$	1	**4**	**12**	0	**4**	0	0	0	**12**	0	0	0	0	0	0	0	**4**

n	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	**33**	34
$a(n) \bmod 16$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	**12**	0

The data fit the conjecture up to $n = 16$, but at $n = 17$ the remainder is 4, not 0.

1.3 What this paper proves

Main results (Corollary 3.5, and Theorems 3.2 and 3.3).

1. *For $n \geq 2$, the remainder of $a(n)$ on division by* 16 *is* 4 *if $n = 4^k + 1$ (that is, $n = 2, 5, 17, 65, \ldots$), it is* 12 *if $n = 2 \cdot 4^k + 1$ ($n = 3, 9, 33, 129, \ldots$), and it is* 0 *for all other n.*

   So the first half of the conjecture is true: for $n > 8$, the remainder is 12 exactly when $n = 2 \cdot 4^k + 1$. The second half, "otherwise $a(n)$ is divisible by 16", is false: it fails at $n = 17, 65, 257, \ldots$, where the remainder is 4. The smallest counterexample is $a(17) = 113101684573952804$. These exceptions are rare (below 1000 there are only three of them after $n = 8$), which may explain why they were missed.

2. *$a(n)$ is divisible by $4^{s_2(n-1)}$, where $s_2(n-1)$ is the number of ones in the binary expansion of $n-1$.* The same holds for the coefficients of the compositional inverse of A.

All of these statements are special cases of theorems about the whole family of equations

$$A_m(x\,A_m(x) - m\,x\,A_m(x)^2) = x^2,$$

in which 4 is replaced by an arbitrary integer m. Hanna's entries A265940, A271958, A396843, A396844 and A396845 [1, 2, 3, 4, 5] are the cases $m = 1, 2, 3, 4, 5$. The heart of the matter is a simple formula for the whole series modulo m^2 (Theorem 3.2):

$$A_m(x) \equiv x + m\,(x^2 - x^3 + x^5 - x^9 + x^{17} - x^{33} + \cdots) \pmod{m^2},$$

where the exponents are the numbers $2^j + 1$ and the signs alternate.

1.4 What was known

The entry records several formulas due to Hanna. One of them, $B(x^2) = x\,A(x) - 4\,x\,A(x)^2$, where B is the compositional inverse of A (Section 2), is the starting point of our proof of the divisibility result. As of October 2026 the entry does not mention the exceptions at $n = 4^k + 1$. We are not aware of any other proof of these statements.

1.5 Why it is interesting

Equation (1) mixes two operations, multiplication and substitution. Such equations rarely have explicit solutions: the coefficients can be computed one after another, but there is usually no formula for them, and they grow quickly. Hanna has contributed many sequences of this kind to the OEIS. For such sequences, divisibility patterns are often the only structure that is visible in the data, and proving them is one way to understand the sequence.

The divisibility result is perhaps the most surprising of the three. Statements in which the binary digits of n control divisibility are classical for explicit numbers. For instance, Legendre's formula says that the exponent of 2 in $n!$ equals $n - s_2(n)$, where $s_2(n)$ is the number of ones in the binary expansion of n [6]. Here the same kind of structure appears in a series that is defined only implicitly.

Finally, the proofs use a few short ideas that recur throughout this collection of papers: reduce the equation modulo a small number, guess the solution, and conclude by uniqueness; and use the compositional inverse to turn an implicit equation into a recursion.

1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution, compositional inverses, congruences, binary digits and prime exponents. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the results and open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed. Standard references for formal power series are [9, 8].

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1+x+x^2+\cdots$ is the *multiplicative inverse* of $1-x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of U below degree d are zero. So $U - V = O(x^d)$ means that U and V agree up to degree $d-1$. A series of the form $x + u_2 x^2 + u_3 x^3 + \cdots$, with constant term 0 and coefficient of x equal to 1, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1 V(x) + u_2 V(x)^2 + u_3 V(x)^3 + \cdots.$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \geq 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. For example, $17 \equiv 5 \pmod{4}$, since both leave remainder 1, and $-3 \equiv 1 \pmod{4}$. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$. Saying that a is odd is the same as saying $a \equiv 1 \pmod{2}$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Several lemmas below hold over every *commutative ring*, that is, every set with an addition, subtraction and multiplication satisfying the usual rules of arithmetic (for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$). No division is used; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The method

Almost every proof in this paper follows the same three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/m\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo m gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A good guess.* Find a simple series that satisfies the reduced equation and check that it does. By uniqueness, this simple series *is* the reduction of the sequence, and its coefficients can simply be read off.

The creative part is step 3: finding a simple series that works. Step 1 sometimes needs care, when the equation seems to require a division. For a first example, take the Catalan series $C = x + C^2$ from above. Its coefficients $1, 1, 2, 5, 14, 42, \ldots$ are determined one at a time by this equation. Modulo 2, the series $S = x + x^2 + x^4 + x^8 + \cdots$ satisfies the same equation: in

$\mathbb{F}_2$, $S^2 = x^2 + x^4 + x^8 + \cdots$ (see Lemma 2.2), so $x + S^2 = S$. Hence $C \equiv S \pmod 2$, and the Catalan coefficient $[x^n]\,C$ is odd exactly when n is a power of 2.

2.6 Squaring modulo 2

Lemma 2.2 (Frobenius identity)**.** *Let p be a prime. For every power series U with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

Proof. In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by p. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 just doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

2.7 Binary digits

Every whole number $r \geq 0$ can be written in exactly one way as a sum of distinct powers of 2; this is its binary expansion. For example $13 = 8 + 4 + 1$, which is written 1101 in binary. We write $s_2(r)$ for the number of powers of 2 in this sum, that is, the number of ones in the binary expansion of r. So $s_2(0) = 0$, $s_2(13) = 3$, $s_2(2^k) = 1$, and $s_2(2^k - 1) = k$, because $2^k - 1 = 1 + 2 + 4 + \cdots + 2^{k-1}$. We use two simple facts:

$$s_2(2r) = s_2(r), \qquad s_2(r_1 + r_2 + \cdots + r_k) \leq s_2(r_1) + s_2(r_2) + \cdots + s_2(r_k). \tag{2}$$

The first holds because doubling a number shifts its binary digits one place to the left. For the second, write each r_i as a sum of distinct powers of 2 and put all these sums together. This writes the total as a sum of $s_2(r_1) + \cdots + s_2(r_k)$ powers of 2, possibly with repetitions. Now repeatedly replace two equal powers $2^i + 2^i$ by the single power 2^{i+1}. This does not change the value, and it lowers the number of terms by one. When no repetitions are left, we have the binary expansion of the total, with $s_2(r_1 + \cdots + r_k)$ terms.

2.8 Prime exponents

For a prime number p and a nonzero integer z, we write $v_p(z)$ for the exponent of p in the prime factorization of z, that is, the largest e such that p^e divides z. For example $975 = 3 \cdot 5^2 \cdot 13$, so $v_3(975) = 1$ and $v_5(975) = 2$. Every power of p divides 0, so we set $v_p(0) = \infty$.

3 Results

We treat the whole family at once. For an integer m, consider the equation

$$A_m(x\,A_m(x) - m\,x\,A_m(x)^2) = x^2, \qquad A_m(x) = x + (\text{terms of degree} \geq 2). \tag{3}$$

Lemma 4.1 in Section 4 shows that for every integer m it has exactly one solution, and that the solution has integer coefficients. We write $a_n(m)$ for the coefficient of x^n in A_m. The sequence of OEIS A396844 is $a(n) = a_n(4)$.

Theorem 3.1 (Parity)**.** *If m is odd, then $a_n(m)$ is odd for every $n \geq 1$. If m is even, then $a_1(m) = 1$ and $a_n(m)$ is even for every $n \geq 2$.*

Theorem 3.2 (Remainders modulo m^2)**.** *Let $m \geq 2$. Then*

$$A_m(x) \equiv x + m \sum_{j \geq 0} (-1)^j \, x^{2^j+1} = x + m\,(x^2 - x^3 + x^5 - x^9 + x^{17} - \cdots) \pmod{m^2}.$$

In other words, for $n \geq 2$: if $n = 2^j + 1$ for some $j \geq 0$, then $a_n(m) \equiv (-1)^j \, m \pmod{m^2}$; for all other n, $a_n(m)$ is divisible by m^2.

Let B_m be the compositional inverse of A_m (Lemma 2.1), so that $A_m(B_m(x)) = x$, and let $b_n(m)$ be its coefficient of x^n. For example, the entry records $B_4(x) = x - 4x^2 + 4x^3 - 48x^4 + 236x^5 - \cdots$. Recall that $s_2(r)$ is the number of ones in the binary expansion of r.

Theorem 3.3 (Binary digits control divisibility)**.** *Let $m \geq 2$ and $n \geq 1$.*

(a) $m^{\,s_2(n-1)}$ divides both $a_n(m)$ and $b_n(m)$. In particular, m^k divides $a_{2^k}(m)$ for every $k \geq 0$.

(b) The same holds with m replaced by a variable t. Section 4 shows that $a_n(m)$ and $b_n(m)$ are the values at $t = m$ of polynomials $a_n(t)$ and $b_n(t)$ with integer coefficients. These polynomials have no terms of degree less than $s_2(n-1)$ in t.

For example, $a_4(t) = 5t^3 - 2t^2$, and $s_2(3) = 2$. The next table illustrates part (a) for $m = 4$. It compares $s_2(n-1)$ with the exponent $v_2(a(n))$ of the prime 2 in $a(n)$. By Corollary 3.4 below, the second is always at least $2\,s_2(n-1)$. Indices in bold are the powers of 2, $n = 2^k$, where $s_2(n-1) = k$.

n	**1**	**2**	3	**4**	5	6	7	**8**	9	10	11	12	13	14	15	**16**
$s_2(n-1)$	**0**	**1**	1	**2**	1	2	2	**3**	1	2	2	3	2	3	3	**4**
$v_2(a(n))$	**0**	**2**	2	**5**	2	5	4	**9**	2	5	4	8	4	8	6	**15**

n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	**32**
$s_2(n-1)$	1	2	2	3	2	3	3	4	2	3	3	4	3	4	4	**5**
$v_2(a(n))$	2	5	4	8	4	9	6	13	4	8	6	11	6	14	8	**16**

Corollary 3.4 (Where the smallest power of a prime occurs)**.** *Let $m \geq 2$, let p be a prime that divides m, and let $e = v_p(m)$. Then $v_p(a_n(m)) \geq e\,s_2(n-1)$ for every $n \geq 1$. Moreover, for $n \geq 2$,*

$$v_p(a_n(m)) = e \quad \Longleftrightarrow \quad n = 2^j + 1 \text{ for some } j \geq 0,$$

and $v_p(a_n(m)) > e$ for all other $n \geq 2$.

Corollary 3.5 (OEIS A396844)**.** *For $m = 4$, that is, for the sequence $a(n)$ of OEIS A396844:*

(a) $a(1) = 1$, and $a(n)$ is even for every $n \geq 2$;

(b) $4^{s_2(n-1)}$ divides $a(n)$ and $b_n(4)$, for every $n \geq 1$;

(c) for $n \geq 2$,

$$a(n) \equiv \begin{cases} 4 \pmod{16}, & \text{if } n = 4^k + 1 \text{ for some } k \geq 0, \\ 12 \pmod{16}, & \text{if } n = 2 \cdot 4^k + 1 \text{ for some } k \geq 0, \\ 0 \pmod{16}, & \text{otherwise.} \end{cases}$$

Hence, for $n > 8$, the remainder is 12 exactly when $n = 2 \cdot 4^k + 1$, as conjectured; but the remainder is 4, not 0, at $n = 17, 65, 257, \ldots$, so the clause "otherwise $a(n)$ is divisible by 16" is false.

4 Proofs

The idea in brief. Each theorem has its own short idea. *Parity:* modulo 2, the equation is solved by $x/(1+x) = x + x^2 + x^3 + \cdots$ when m is odd, and by x when m is even; since the solution is unique, this is the reduction of A_m. *Remainders modulo m^2:* modulo m the solution is just x, so $A_m = x + m\,C$ for some integer series C. Putting this into the equation and discarding multiples of m^2 leaves the simple linear equation $x\,C(x) + C(x^2) \equiv x^3 \pmod m$, whose solution has nonzero terms only in the degrees $2^j + 1$. *Binary digits:* substituting the inverse series into the equation gives an identity that expresses each coefficient through earlier ones. In that identity, an index is either halved, which does not change its binary digit sum, or split into two parts at the cost of one extra factor m. An induction on n then shows the divisibility.

Lemma 4.1 (Existence and uniqueness)**.** *Let R be a commutative ring and let $m \in R$. There is exactly one series $A = x + a_2x^2 + a_3x^3 + \cdots$ with coefficients in R such that $A(xA - mxA^2) = x^2$. For $R = \mathbb{Z}$ this is the series A_m, and for every $q \geq 2$ the reduction of A_m modulo q is the solution over $\mathbb{Z}/q\mathbb{Z}$.*

Proof. For a series $A = x + a_2x^2 + \cdots$ write $T_A = xA - mxA^2$, which begins with x^2, and $\Psi(A) = A(T_A)$. The key claim is: *if A and A^* agree below degree r, where $r \geq 2$, and $A^* - A = c\,x^r + (\text{higher terms})$, then*

$$\Psi(A^*) - \Psi(A) = c\,x^{r+1} + (\text{higher terms}). \tag{4}$$

In other words, changing the coefficient of x^r by c changes the coefficient of x^{r+1} in Ψ by exactly c, and leaves all lower coefficients alone. To see this, split the difference into two parts,

$$\Psi(A^*) - \Psi(A) = \big(A(T_{A^*}) - A(T_A)\big) + (A^* - A)(T_{A^*}).$$

First, $T_{A^*} - T_A = x(A^* - A)\big(1 - m(A + A^*)\big) = c\,x^{r+1} + (\text{higher terms})$. Next, $A(T_{A^*}) - A(T_A) = (T_{A^*} - T_A) + \sum_{k\geq 2} a_k\,(T_{A^*}^k - T_A^k)$, and for $k \geq 2$ the difference $T_{A^*}^k - T_A^k$ is $T_{A^*} - T_A$ times a sum of products of $k-1$ factors, each beginning with x^2; so it begins in degree $r+3$ or later. Hence $A(T_{A^*}) - A(T_A) = c\,x^{r+1} + (\text{higher terms})$. Finally, $(A^* - A)(T_{A^*})$ begins in degree $2r \geq r + 2$, because $A^* - A$ begins in degree r and T_{A^*} in degree 2. This proves (4).

Uniqueness. If two different solutions existed, let r be the first degree in which they differ. By (4), Ψ of the two would differ in degree $r+1$. But both equal x^2.

Existence. Start with $A = x$. If $\Psi(A) - x^2$ has no terms below degree $r+1$ and its coefficient of x^{r+1} is c, replace A by $A - c\,x^r$. By (4) this removes that coefficient and changes nothing below it. Each step fixes one more coefficient of A for good, because the coefficient of x^d in $\Psi(A)$ depends only on $a_2, \ldots, a_{d-1}$. So the steps build a solution, one coefficient at a time.

No division was used, so the argument works over every commutative ring. Reducing the integer solution modulo q gives a solution over $\mathbb{Z}/q\mathbb{Z}$, because reduction is compatible with all the operations; by uniqueness it is *the* solution there. □

Running the construction with a variable t in place of m, that is, over the ring $\mathbb{Z}[t]$ of polynomials in t with integer coefficients, gives

$$A = x + t\,x^2 + (2t^2 - t)\,x^3 + (5t^3 - 2t^2)\,x^4 + \cdots,$$

and in general a polynomial $a_n(t)$ with integer coefficients as the coefficient of x^n. Substituting the number m for t is compatible with all the operations, so by uniqueness it turns this series into A_m. So $a_n(m)$ is the value of $a_n(t)$ at $t = m$; for example $a_3(3) = 2 \cdot 9 - 3 = 15$. The same applies to the inverse series and $b_n(t)$.

Proof of Theorem 3.1. Let m be odd and reduce modulo 2. Since $m \equiv 1$ and $-1 \equiv 1$, the equation becomes $P(xP + xP^2) = x^2$ over $\mathbb{F}_2$. We try $H = x/(1+x) = x + x^2 + x^3 + \cdots$. Working in $\mathbb{F}_2[[x]]$, where $(1+x)^2 = 1 + x^2$ (Lemma 2.2),

$$xH + xH^2 = \frac{x^2(1+x) + x^3}{(1+x)^2} = \frac{x^2 + 2x^3}{(1+x)^2} = \frac{x^2}{(1+x)^2},$$
$$H\left(\frac{x^2}{(1+x)^2}\right) = \frac{x^2}{(1+x)^2 + x^2} = \frac{x^2}{1 + 2x + 2x^2} = x^2,$$

because $2 = 0$ in $\mathbb{F}_2$. (For the second line we used $H(y) = y/(1+y)$ and multiplied the numerator and the denominator by $(1+x)^2$.) So H solves the reduced equation. By Lemma 4.1, $A_m \equiv H \pmod 2$: all coefficients of positive degree are odd.

If m is even, the reduced equation is $P(xP) = x^2$. It is solved by $P = x$, because $x \cdot x = x^2$. So $A_m \equiv x \pmod 2$, which is the claim. □

Proof of Theorem 3.2. Step 1: modulo m. Modulo m the equation becomes $P(xP) = x^2$, which is solved by $P = x$. By Lemma 4.1, $A_m \equiv x \pmod m$. So every coefficient of $A_m - x$ is divisible by m, and we can write $A_m = x + m\,C$ for a series C with integer coefficients that begins in degree 2.

Step 2: modulo m^2. Substitute $A_m = x + mC$ into $T = xA_m - mxA_m^2$:

$$\begin{aligned} T &= x^2 + m\,xC - m\,x\,(x + mC)^2 = x^2 + m\,(xC - x^3) - 2m^2x^2C - m^3xC^2 \\ &\equiv x^2 + m\,(xC - x^3) \pmod{m^2}. \end{aligned}$$

Since $T \equiv x^2 \pmod m$, also $C(T) \equiv C(x^2) \pmod m$, and so $m\,C(T) \equiv m\,C(x^2) \pmod{m^2}$. Therefore

$$x^2 = A_m(T) = T + m\,C(T) \equiv x^2 + m\,(xC(x) - x^3 + C(x^2)) \pmod{m^2}.$$

So every coefficient of $m(xC(x) + C(x^2) - x^3)$ is divisible by m^2. Dividing by m, this says

$$x\,C(x) + C(x^2) \equiv x^3 \pmod m. \tag{5}$$

Step 3: solving (5). The series $S = \sum_{j\ge0}(-1)^j x^{2^j+1} = x^2 - x^3 + x^5 - x^9 + \cdots$ satisfies (5) exactly, because

$$x\,S = x^3 - x^4 + x^6 - x^{10} + x^{18} - \cdots, \qquad S(x^2) = x^4 - x^6 + x^{10} - x^{18} + \cdots,$$

and everything except x^3 cancels. Moreover, (5) has only one solution over $\mathbb{Z}/m\mathbb{Z}$ that begins in degree 2. Indeed, suppose two solutions differed, and their difference D began in degree $d \ge 2$. Then xD would begin in degree $d+1$, while $D(x^2)$ begins in degree $2d > d+1$. So $xD + D(x^2)$ would have a nonzero coefficient in degree $d+1$, although it must be 0. Hence $C \equiv S \pmod m$, and $A_m = x + mC \equiv x + mS \pmod{m^2}$. □

Proof of Theorem 3.3. Divide both series by x: write $A_m = x\,F$ and $B_m = x\,G$ with

$$F = f_0 + f_1x + f_2x^2 + \cdots, \qquad G = g_0 + g_1x + g_2x^2 + \cdots, \qquad f_0 = g_0 = 1.$$

Then $f_r = a_{r+1}(m)$ and $g_r = b_{r+1}(m)$, and we must show that $m^{s_2(r)}$ divides f_r and g_r.

Two identities. Substitute both sides of (3) into B_m. Since $B_m(A_m(y)) = y$ for every series y without constant term, this gives $xA_m - mxA_m^2 = B_m(x^2)$, which is Hanna's formula. In terms of F and G it reads $x^2F - mx^3F^2 = x^2G(x^2)$, that is,

$$F(x) = G(x^2) + m\,x\,F(x)^2, \qquad \text{so} \qquad f_r = [\,r \text{ even}\,]\,g_{r/2} + m\sum_{i+j=r-1} f_if_j \quad (r \ge 1), \tag{6}$$

where $[\,r \text{ even}\,]$ is 1 if r is even and 0 otherwise. The second identity is $A_m(B_m(x)) = x$. Since $A_m(y) = \sum_j f_j y^{j+1}$, it reads $\sum_{j\ge 0} f_j x^{j+1} G^{j+1} = x$, that is, $\sum_j f_j x^j G^{j+1} = 1$. Its coefficient of x^r, for $r \ge 1$, gives

$$g_r = -f_r - \sum_{j=1}^{r-1} f_j\,[x^{r-j}]\,G^{\,j+1}. \tag{7}$$

The induction. We show that $m^{s_2(r)}$ divides f_r and g_r, by induction on r. For $r = 0$ there is nothing to show, because $s_2(0) = 0$. Let $r \ge 1$, and assume the claim for all smaller indices. We first treat f_r, using (6). If r is even, $g_{r/2}$ is divisible by $m^{s_2(r/2)} = m^{s_2(r)}$, by (2). Each term $m\,f_i f_j$ with $i+j = r-1$ is divisible by $m^{1+s_2(i)+s_2(j)}$, and $1+s_2(i)+s_2(j) \ge s_2(1+i+j) = s_2(r)$, again by (2) (note $s_2(1) = 1$). So $m^{s_2(r)}$ divides f_r.

Now we treat g_r, using (7). The term f_r was just handled. Each remaining term $f_j\,[x^{r-j}]\,G^{j+1}$ is a sum of products $f_j\,g_{i_1} g_{i_2} \cdots g_{i_{j+1}}$ with $j + i_1 + \cdots + i_{j+1} = r$ and all indices smaller than r. By the induction hypothesis such a product is divisible by $m^{s_2(j)+s_2(i_1)+\cdots+s_2(i_{j+1})}$, and this exponent is at least $s_2(r)$ by (2). So $m^{s_2(r)}$ divides g_r, which completes the induction and proves (a). For the special case, note $s_2(2^k - 1) = k$.

For (b), run the same argument over the ring $\mathbb{Z}[t]$, with the variable t in place of m. Nothing changes: the induction only uses the identities (6) and (7) and the fact that divisibility by powers of t is preserved by sums and products. □

Proof of Corollary 3.4. Since p^e divides m, Theorem 3.3 shows that $p^{e\,s_2(n-1)}$ divides $a_n(m)$, which is the lower bound. Now let $n \ge 2$. If $n = 2^j + 1$, Theorem 3.2 gives $a_n(m) = m\big((-1)^j + m\,q\big)$ for some integer q. The second factor is ± 1 plus a multiple of p, so it is not divisible by p; hence $v_p(a_n(m)) = v_p(m) = e$. If $n \ge 2$ is not of this form, then $n - 1 \ge 1$ is not a power of 2, so $s_2(n-1) \ge 2$, and the lower bound gives $v_p(a_n(m)) \ge 2e > e$. □

Proof of Corollary 3.5. Take $m = 4$ in Theorems 3.1, 3.2 and 3.3. In Theorem 3.2, an index $n = 2^j + 1$ has either $j = 2k$ even, so that $n = 4^k + 1$ and the remainder is $4 \cdot (+1) = 4$, or $j = 2k + 1$ odd, so that $n = 2 \cdot 4^k + 1$ and the remainder is $4 \cdot (-1) = -4 \equiv 12 \pmod{16}$. The indices $4^k + 1$ with $k \ge 2$ are larger than 8 and are not of the form $2 \cdot 4^k + 1$, so they contradict the conjectured clause. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We computed A with the construction recorded in the entry: starting from $A = x$, repeatedly replace A by $(1 - \sqrt{1 - 16B(x^2)/x})/8$, where B is the compositional inverse of the current A. This reproduces all 22 terms printed in the entry, and the result satisfies the defining equation (1) through degree 260. For every $n < 260$, the remainders modulo 16 are as in Corollary 3.5, $a(n)$ is even for $n \ge 2$, and $4^{s_2(n-1)}$ divides both $a(n)$ and $b_n(4)$. In particular $a(17) \equiv 4 \pmod{16}$. For the general family, Theorems 3.1, 3.2 and 3.3(a) and the characterization in Corollary 3.4 hold for every m from 2 to 12 and every $n < 70$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [12]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [10] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [11], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A396844.lean` of the repository, in the namespace `OEIS.A396844`. The general theorems of Section 3, for every m, are proved in the companion file `lean/Entries/A396843.lean`; see the paper on A396843 for the corresponding table. The failure of the conjectured clause at $n = 17$ is the case $k = 2$ of the first part of `residues_parameter_four`. The table lists where each result of this paper is proved.

Statement in this paper	Lean theorem(s)
Lemma 4.1 for $m = 4$, over every commutative ring	`source_existence_uniqueness`
Corollary 3.5(a)	`even_positive_tail`
Corollary 3.5(b), for $a(n)$	`binary_divisibility`
Corollary 3.5(c): remainders 4, 12 and 0 modulo 16	`residues_parameter_four`

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

6 Discussion

Scope. The results replace the conjecture of the entry by a complete and proved description of the remainders modulo 16, and they show exactly where the conjecture fails. They say nothing about the size of the coefficients, or about the radius of convergence that the entry also discusses.

Higher powers of m. Theorem 3.3 also gives information modulo every power of m: modulo m^k, the coefficient $a_n(m)$ is 0 unless $n-1$ has fewer than k ones in its binary expansion. So modulo m^k the nonzero remainders can occur only at a very sparse set of indices. For $k = 2$, Theorem 3.2 describes these remainders completely. For $k = 3$ the candidates are $n = 1$, $n = 2^i + 1$ and $n = 2^i + 2^j + 1$ with $i > j$. We did not try to describe the remainders modulo m^3; this seems a natural next question.

Other values of m. For $m = 1$ (entry A265940) the results say only that every coefficient is odd. For $m = 2$ (entry A271958) they say, for example, that $a_n(2) \equiv \pm 2 \pmod 4$ when $n = 2^j + 1$ and $a_n(2) \equiv 0 \pmod 4$ for the other $n \geq 2$. For $m = 5$ (entry A396845) the remainders modulo 25 are 5 and 20 at $n = 4^k + 1$ and $n = 2 \cdot 4^k + 1$, and 0 elsewhere.

Why the exceptions are easy to miss. The exceptional indices $n = 4^k + 1$ grow exponentially, and the conjecture excluded $n \leq 8$. So the first index at which the conjecture can be tested against an exception is $n = 17$, and the next is $n = 65$. A table with fewer than 17 terms cannot reveal the problem. This is a small example of why a proof, rather than more data, is needed to settle such a pattern.

About this paper

How the work was done. This paper is one of 26 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [12]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [12], where the Lean sources and the numerical checks are also available.

Acknowledgements

The family of sequences and the conjectures are due to Paul D. Hanna [1, 2, 3, 4, 5]. The identity $xA - mxA^2 = B(x^2)$, used in the proof of Theorem 3.3, is recorded in his entries.

References

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